

Finals Practice Task 01. Data Prep And EDA using SQL

Description:

This task introduced the use of MySQL Workbench for basic data cleaning and preparation. Using the dirty_shipments dataset, different issues such as incorrect formatting, missing or negative values, and inconsistent data entries were identified and corrected. Additional columns were also created to calculate the difference between shipment dates and delivery dates. The activity helped improve understanding of data organization, cleaning techniques, and the importance of maintaining accurate and reliable datasets before analysis.

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'Schemas' panel with a tree view of the database structure, including tables like 'shipment_id', 'origin_warehouse', 'destination_city', 'carrier', 'shipment_status', 'freight_cost', 'items_count', 'damage_reported', 'delivery_date', 'ship_date', 'total_days', 'delivery_flag', 'weight_valid', and 'row_num'. The main editor window contains a SQL query that uses a window function to partition data by origin_warehouse, destination_city, carrier, ship_date, and CONVERT(weight_kg, char), and then orders by shipment_status and shipment_id. The query is limited to 50 rows. Below the query editor, the 'Result Grid' shows a table with columns: shipment_id, origin_warehouse, destination_city, destination_state, carrier, shipment_status, freight_cost, items_count, damage_reported, delivery_date, ship_date, total_days, delivery_flag, weight_valid, and row_num. The table contains 10 rows of data. The bottom panel shows the 'Procedure: cleaned_shipment_query' with an 'Action Output' tab displaying a list of actions and their results, including a message about an unknown column 'row_num' in the 'where clause'.

```
select shipment_id, row_number() over (
partition by
origin_warehouse,
destination_city,
destination_state,
carrier,
ship_date,
CONVERT(weight_kg, char),
CONVERT(freight_cost, char),
shipment_status,
damage_reported
ORDER BY shipment_id
) as temp
from shipment_tbl
) as temp
WHERE row_num = 2
```

shipment_id	origin_warehouse	destination_city	destination_state	carrier	shipment_status	freight_cost	items_count	damage_reported	delivery_date	ship_date	total_days	delivery_flag	weight_valid	row_num
SP-1003	Warehouse A	Dallas	TX	Fastfreight	Delivered	653	45	No	2024-01-31	2024-01-30	4	Valid	120.5	2
SP-1017	Warehouse C	Denver	CO	Quickship	Pending	0	0	Yes	2024-02-23	2024-02-18	5	Valid	0	1
SP-1007	Warehouse D	Seattle	WA	Quickship	Delivered	980	110	Yes	2024-01-25	2024-01-20	5	Valid	210.8	1
SP-1010	Warehouse A	Dallas	TX	Fastfreight	Delivered	390	65	No	2024-02-05	2024-02-01	4	Valid	175.3	1
SP-1019	Warehouse A	Atlanta	GA	Fastfreight	In Transit	540	60	No	2024-01-14	2024-02-22	38	Invalid	165	1
SP-1001	Warehouse A	Dallas	TX	Fastfreight	Delivered	490	48	No	2024-01-14	2024-01-10	4	Valid	120.3	1
SP-1023	Warehouse A	Dallas	TX	Fastfreight	Delivered	490	48	Yes	2024-02-09	2024-03-05	4	Valid	120.3	1
SP-1015	Warehouse A	Houston	TX	Fastfreight	Delivered	370	40	No	2024-02-16	2024-02-12	4	Valid	98.4	1
SP-1005	Warehouse A	Houston	TX	Quickship	Delivered	280	20	Yes	2024-01-20	2024-01-15	5	Valid	45.2	1